

Optimizing Inventory Management And Improving Profit of Textile Shop To Elevate Business Efficiency Using Data Analysis

The Final Report For BDM Capstone Project

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EXECUTIVE SUMMARY

I have been working with Mr.Mohamed Abbas Ali for this project who runs a textile business named ‘Namnadu silks ‘ located at Sirkali,Mayiladuthurai,Tamil Nadu. This is a B2C business . He sells a wide range of products for Men and Women.

The owner is facing significant challenges in increasing the profit margins and managing inventory. He struggles to monitor inventory levels and he is unable to identify the items that are overstocked or running low . The overstocking of items will affect its profitability even though it performs well in sales.The Business expenses are considerable and showing minor fluctuations , but the profit has fallen short of expectation and has been declining for the past few months.This affects the net profit . The reduction in net income makes it difficult for him to invest towards the business’s growth .

I have gathered 6 months data and calculated different key metrics including Days sales of inventory,stock-sales ratio,product contribution,expenses-sales ratio,net sales,net profit for better understanding of the trends in the purchase,sales,inventory and expense data.I have utilized data visualization tools like bar chart,column chart,pie chart to analyze the metrics and extract valuable insights.

I have given data driven suggestions for sales,inventories and purchase . Focusing more on the sales on high contributing items , regularly reviewing the slow moving items inventories to avoid over stocking and fast moving items inventories to avoid running out of stock will help to increase the profit .Implementing different techniques can lead to significant growth in business.

DETAILED EXPLANATION OF ANALYSIS PROCESS/METHOD

The objective of this analysis is to identify and understand the issues and trends in inventory management,sales,purchase and profitability and aim to provide data driven recommendations for improvement.

After talking about the problems I have collected data about sales,purchase,inventory and expenses for 6 months. I have cleaned the data ,verified the data consistency,removed the duplicates in the data. They had provided me with the purchase data for 6 months. . They hadn't purchased every item monthly . For the convenience of calculation and understanding , I have organized the data month wise and entered the purchase details for the items in the particular month and entered 0 for the product details which aren't purchased in the particular month.

To perform analysis and get insights , I have already calculated gross profit ,cost of goods sold,inventory turnover,sell through rate . I am calculating several other key metrics to understand the data furthermore. I have used microsoft excel for calculations and charts.

Inventory Data :

Analyzing inventory will give the measure of stock management,operational efficiency and its impact on profitability. As from previous analysis it is understandable that the stocks aren't managed properly . Some stocks had been overstocked and there were chances of stock shortage.

I have found Days Sales of inventory for each product to track the trend and efficiency of inventory management . This will be useful to find the areas which need to be improved.

$$DSI = (\text{Average Inventory} / \text{cost of goods sold}) * (\text{Time period})$$

By calculating the stock sales ratio , it is helpful to measure the relationship between the amount of inventory held by the business and sales made in the specified period.

$$SSR = (\text{Average Inventory} / \text{Net sales}) * 100$$

I also calculated Gross margin return on investment . It measures the rate of profit made from its inventory .

$$GMROI = \text{gross profit} / \text{average inventory}$$

Sales Data :

Analyzing sales data will help to identify trends , profit, and areas which need improvement.

Sales also determine the level of inventory. The sales of the products will help to understand the demands.

I have already calculated gross sales(selling price) , to find net sales I have discussed it with the owner about it . In the previous analysis I found out that baby items and men's unstitched fabric had been sold with loss. He mentioned that he had sold both the products with discounts and also he clearly said that they won't sell any damaged products to the customers and strictly follows a No return policy .

$$\text{NET SALES} = \text{Gross sales} - \text{Sales Returns} - \text{Sales Allowance} - \text{Sales Discounts}$$

He said there was no amount spent on sales returns and sales allowances but he had put a discount for 2 products in this 6 months.

Month - wise product contribution makes it easier to observe the trends of the products across different months .

$$\text{PRODUCT CONTRIBUTION} = \left(\frac{\text{Sales of the product in the month}}{\text{total sales in the month}} \right) * 100$$

Purchase Data :

Analyzing the purchase data helps to improve the cost efficiency, profitability and inventory planning.

I am calculating month wise product contribution for each product to observe the purchase trend of product across different month

$$\text{PRODUCT CONTRIBUTION} = \left(\frac{\text{Purchase cost of product}}{\text{total purchase cost in the month}} \right) * 100$$

Purchase-sales ratio is a valuable measure used for inventory management and demand forecasting. It is helpful to understand the balance between how much stock is purchased and sold .

$$PSR = (\text{Total purchase} / \text{Total sales}) * 100$$

Expense Data :

Apart from purchase cost , there are several expenses for this textile shop . Analyzing the expense data will help me to understand the profitability and financial condition of this business

Expense to Sales Ratio is an important financial measure which compares the company's expenses with its revenue . It is useful to observe the proportion of revenue used for the other expense purpose which includes electricity bill,rent,maintenance charge,salary.

$$ESR = (\text{Total Expense} / \text{Total revenue}) * 100$$

MONTH	ESR VALUE
March	16.52
April	11.07
May	10.03
June	13.16
July	14.07
August	14.26

Table-1 Month Wise Expense-Sales Ratio

The month March has the highest ESR value which indicates higher expenses relative to the revenue generated in that month. The Lowest ESR value in May shows that the smaller proportion of sales was spent on expenses . The major reason behind these fluctuations in ESR is variation in revenue generated . The expenses for the 6 months show minimal variation , but there was a noticeable difference in the total revenue generated by each month. In this case The increase in ESR in march,june,july,august denotes the decrease in the revenue during those

months . This suggests the business is operating with decreasing revenue and if the expenses aren't reduced , it could affect the net profit .

The net profit earned by the business during those six month is

$$\begin{aligned}\text{NET PROFIT} &= \text{Gross Profit} - \text{Other Expenses} \\ \text{NET PROFIT} &= ₹18,67,820.75 - ₹9,59,436.75 \\ &= ₹ 9,083,84\end{aligned}$$

$$\begin{aligned}\text{NET PROFIT MARGIN} &= (\text{Net profit}/\text{total revenue})*100 \\ &= (₹ 9,083,84/₹ 74,85,642.75)*100 \\ &= 12.14 \%\end{aligned}$$

Net profit Margin tells the percentage of revenue remaining as the profit after deducting all expenses .

Correlation values between different variables helps to find the relationship between them.It suggests whether it is positively or negatively related to each other.

Dataset :  Dataset.xlsx

RESULTS AND FINDINGS

Sales Data :

I have performed analysis on the sales data which contains the sales information like quantity sold and price of 11 products for march - august. From the gross profit analysis , I found that fancy & synthetic saree and silk saree have the highest gross profit margin with 31.03 %.

I have used a bar chart to show product-wise net sales.The Net sales provide much more clearer information about the income then gross sales . It helps to understand discounts , allowances and returns .

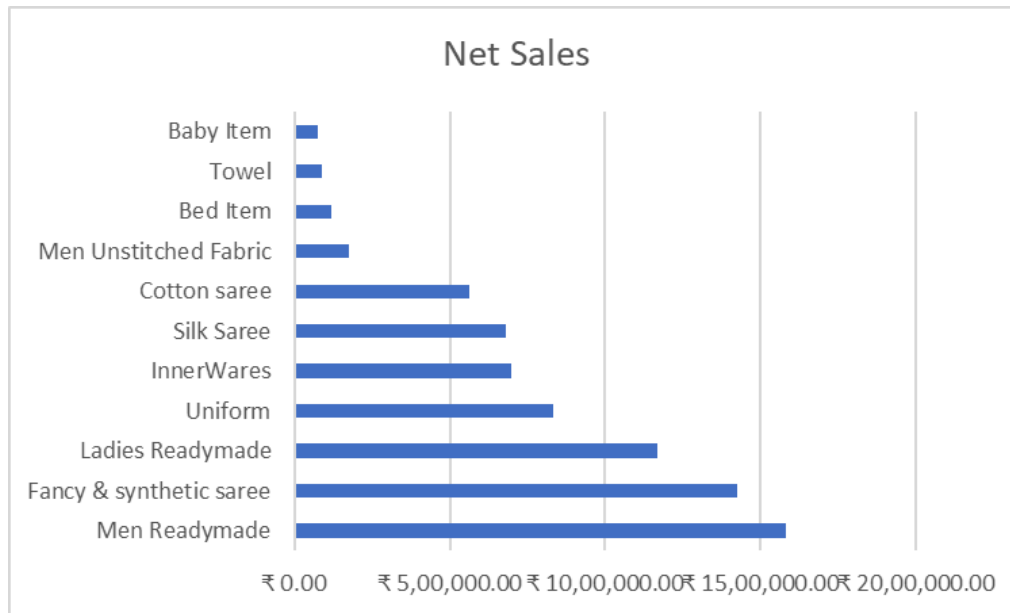
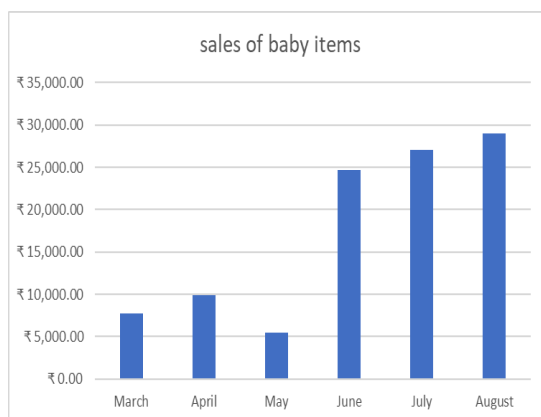


Chart-1 Net sales of the 11 products for March-August

The baby item has the lowest net sales of ₹74,008.00. The sales were minimum for baby items in March, April and May. Then there was a huge incline in sales in the months of June, July and August.



Char-2 Gross sales of baby items for 6 months

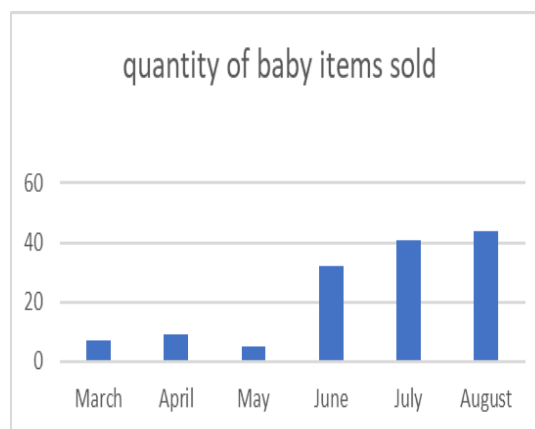


Chart-3 Quantities of baby items sold in each month

It is because they have discounted the baby items in June with 30 % and July, August with 40 % Which inturn had caused 32.9% loss.

The net sales of men's unstitched fabric is ₹ 1,74,106. They have also offered 20 % discount in March and April, 30% discount in May, June, July, August. It caused a 9.03 % loss.

The bed items,towel,uniform,cotton saree,innerwares,cotton sarees,silk sarees have moderate net sales price and Fancy&synthetic saree , ladies readymade have relatively high net sales price.Men's readymade has the highest net sales price of ₹15,81,484.80.

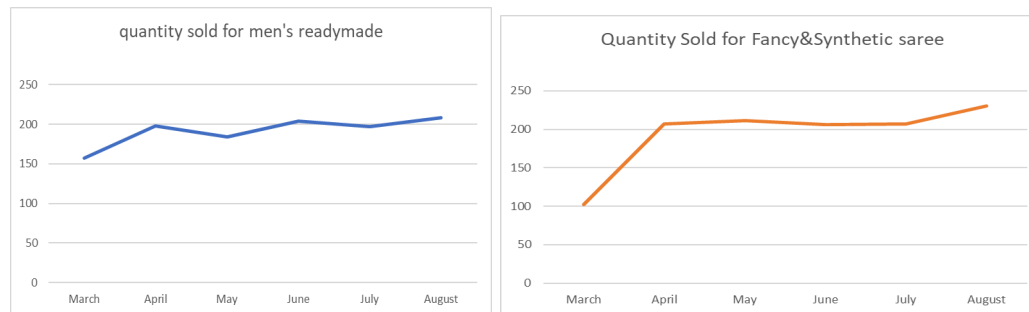


Chart-4 & 5 Quantities sold for men's readymade and Fancy&Synthetic in March-August

The men's readymade and Fancy&synthetic saree sales have been stable with minor fluctuation except the noticeable growth from march to april.

I have used clustered column chart to shows the contribution of each product in sales for march - august.

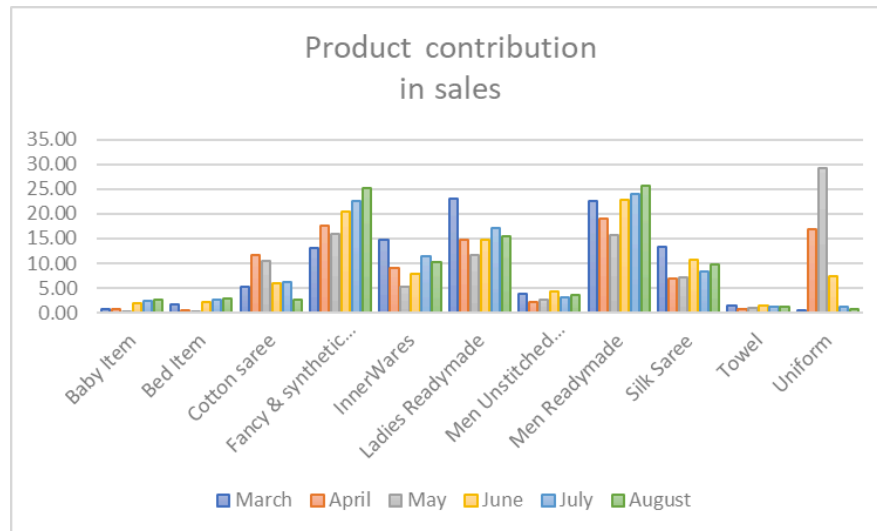


Chart-6 Product wise contribution in sales for each month

The fancy & synthetic saree and men's readymade shows higher contribution across all the months . The sharp increase in the sale of uniforms in May indicated the seasonal demand (for the school reopening purpose. Innerwares, cotton sarees, silk sarees, ladies readymade shows fluctuation in sales. Baby items , bed items and towels have a lower contribution in sales .

Purchase Data :

Analysis on purchase data include purchasing of products across months, product wise production in purchasing across different months.

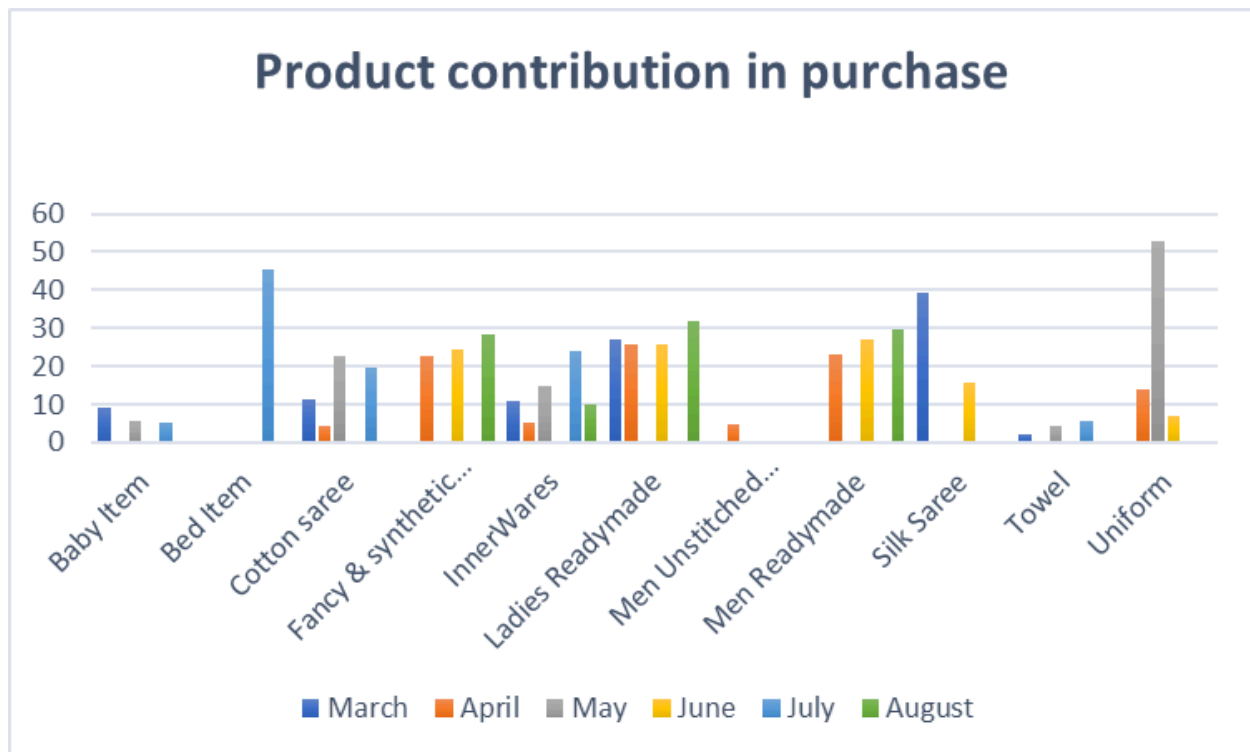


Chart-7 Product wise contribution in purchase for each month

The contribution of ladies readymade in purchase is higher than others . It was purchased in March ,April , June, August and contributes 26.81%,25.80%, 25.83%,31.92% to the total purchase of the respective month. men's unstitched fabric was bought less frequently and contributed only 4.82 % in 6 months . Fancy & Synthetic and men readymade indicates a moderate contribution in purchase .

The below line chart denotes the trends of quantity of ladies readymade purchased and sold.

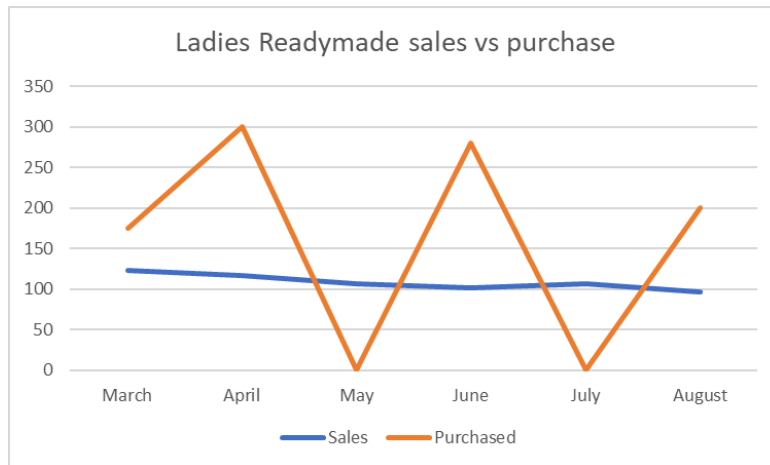


Chart-8 correlation between sales and purchase of ladies readymade

The purchase contribution of ladies readymade is higher than all other products but the sales contribution of ladies readymade isn't relatively high. In the month of May and July no purchase was made for ladies readymade. The chart shows the sales of the product doesn't correlate with the purchase. When the correlation coefficient is calculated between the two variables it results as 0.106. This indicates a very weak positive relationship between the sales and purchase of ladies readymades. This clearly shows that ladies readymades inventories aren't properly managed and overstocked in the inventories.

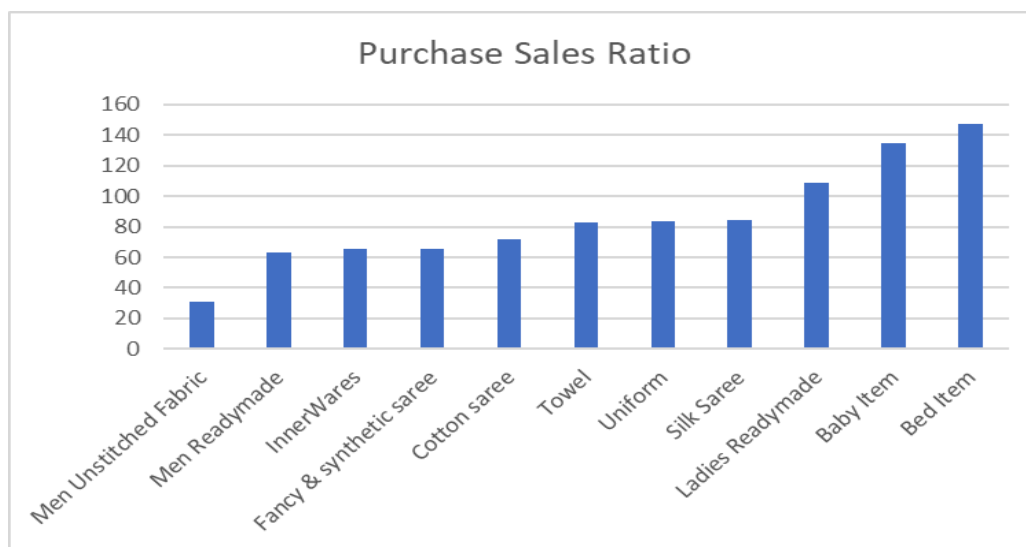


Chart-9 Purchase sales ratio for each products for 6 months

This chart shows that Fancy & synthetic sarees ,innerwares,Men’s readymades and Men unstitched fabric are purchased and sold in a balanced manner for March - August. The purchase-sales ratio of ladies readymade,baby item,bed item was high which indicates that the stocks are excess in inventories

Inventory Data :

Analyzing the inventory data is very important as the poor management of inventory directly affects the profitability of business .

The below column chart shows the days sales of inventory for all the product’s



Chart-10 DSI values of 11 products

The products with higher DSI value are Bed items with 313.02 , Men unstitched fabric 178.74,silk sarees 103.22 . These products are slow moving products. The very high value of Bed items shows that the product isn’t sold in the greater proportion. Though Men unstitched fabric has balanced sales-purchase relationship for march -august, it was overstocked before march . Only by offering discounts, they were able to reduce the stocks .

Silk saree generated a highest gross profit of ₹ 2,10,330 with a gross profit margin of 31.30 % but it was a slow moving product. This is because the silk saree is costlier . Even when the minimal quantity is sold , it can generate a high revenue .

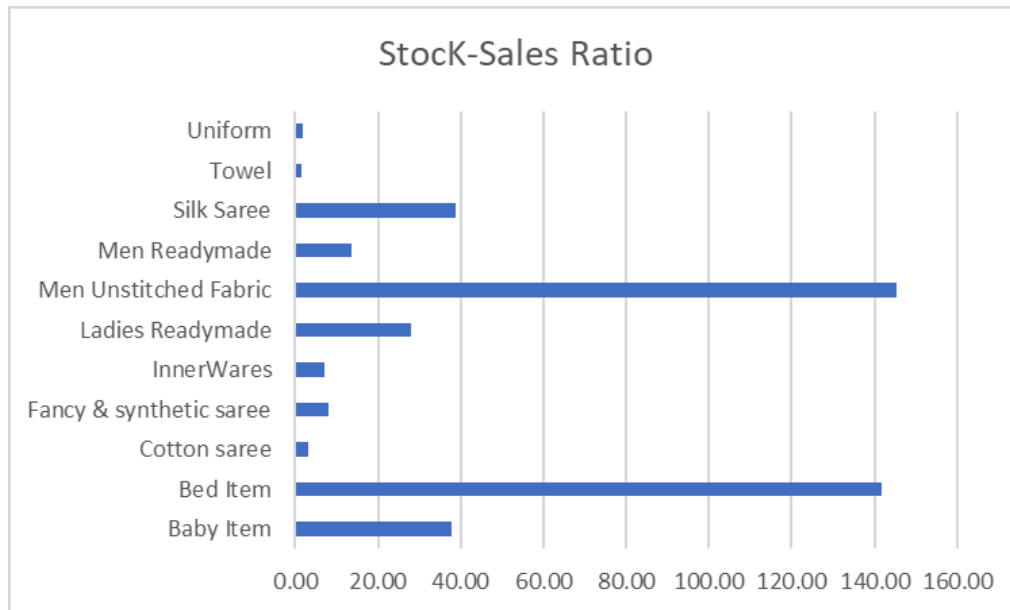


Chart-11 bar chart shows the stock sales ratio

With respect to the sales, the stock values of men's unstitched fabric, silk sarees, bed items, baby items and ladies readymades are high. These five products had been overly stocked. The stock levels Men readymade, innerwares, Fancy & synthetic sarees are maintained in balance with the sales level. The stock level of cotton saree, uniform, and towel are under-stocked. If the demand increases for these products, then it might run out of stock.

The correlation value between the Average inventory level and gross profit of silk saree is -0.82. It denotes a strong negative linear relationship. It signifies a strong inverse relationship when the inventory level increases, the gross profit decreases.

Chart on GMROI :

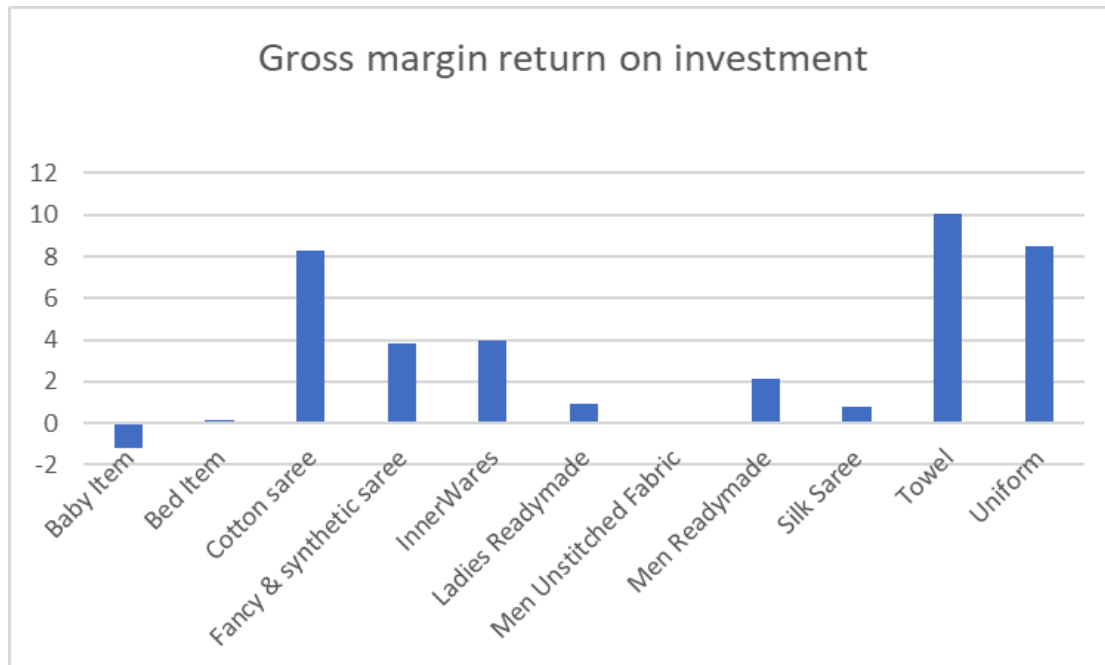


Chart -12 column chart shows the GMROI values for each products across 6 months

The Gross Margin return on investment is high for cotton saree , towel,uniform with 3.82,10.04,8.52. As these products are moving products and not stocked overly, there is lesser negative impact on profit because of these products' inventories . But as stated before if the demand increases the stock level won't be enough to meet the demand .

Fancy & synthetic saree , Men readymade, innerware shows moderate profitability and it has enough stock levels to face demands.

Ladies readymade,silk sarees,bed items have lower values which indicates much profit isn't generated due to the stocks in the inventories .

As i already mentioned baby items(-1.22) and men's unstitched fabric(-0.09) has been overly stocked and has negative GMROI indicates losing money in the inventories

The inventory is negatively correlated with the profit. The more unwanted inventories affect the net profit.

The below pie charts shows the percentage of contribution of product's to profit and loss

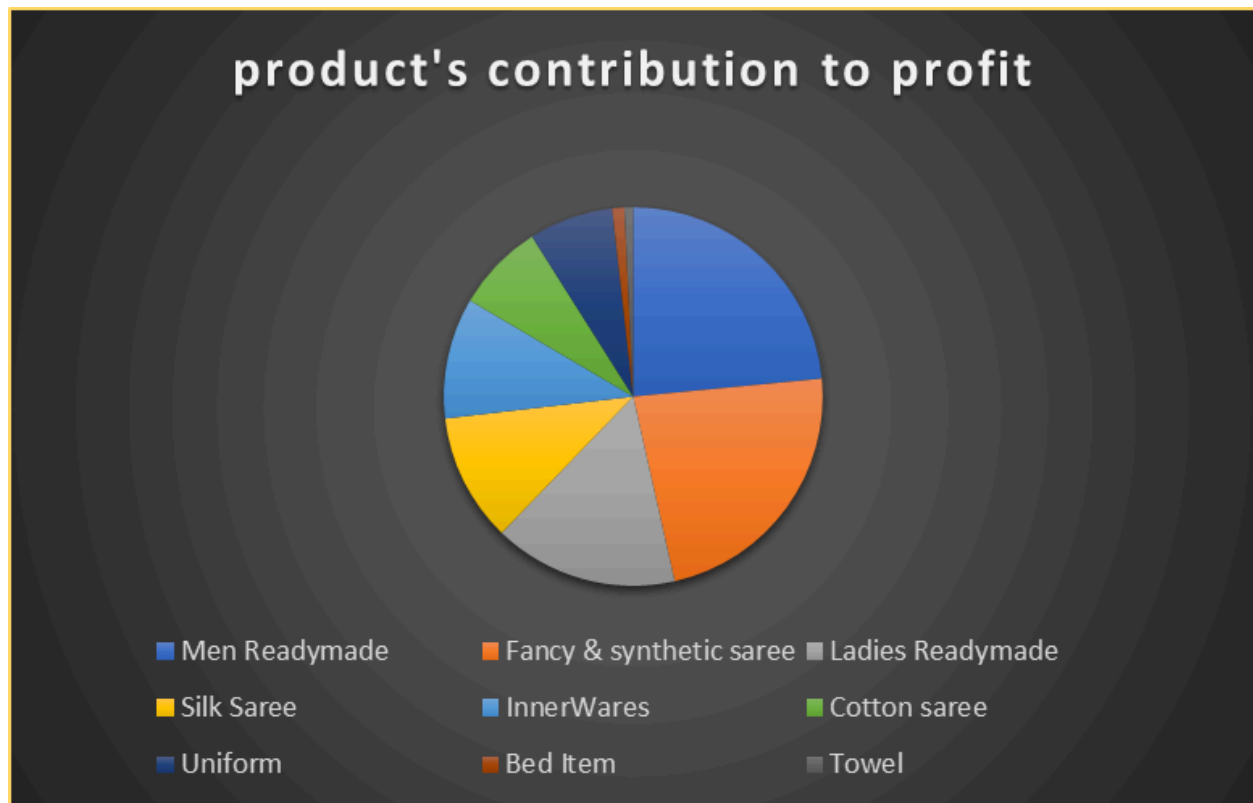


Chart-13 profit percentage of each products

The overall profit (ignoring loss) for the months March to August is ₹19,23,585.75. The Men readymade shares the highest profit percentage (23.49) generating ₹ 4,58,852.80 gross profit. The fancy & synthetic shares 22.99 % of contribution with ₹4,42,230.75. Ladies readymade ,silk saree,Innerwares share a moderate profit of 15.73%,10.93%,10.33% with ₹3,02,584.80,₹2,10,330.00,₹1,98,722.00 gross profit. The cotton saree , uniform sarees contribute 7.55% and 7.20 % . Towels have the lowest profit . It contributes 0.74% with ₹14,310 gross profit.

The overall loss in March - August is ₹55,765.00.

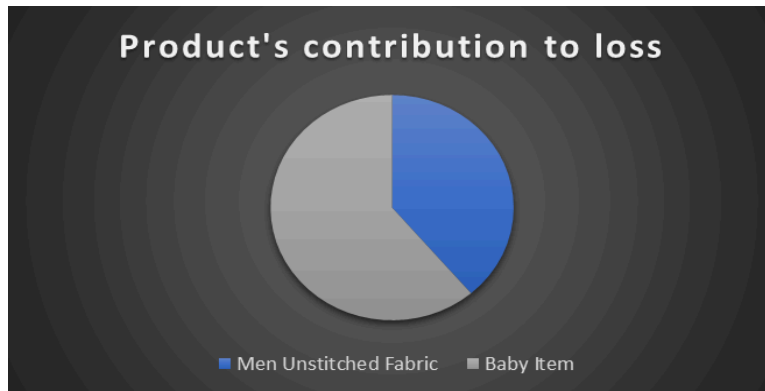


Chart-14 Percentage of loss

The baby item has the highest loss of ₹34,160 (61.26%). The men's Unstitched Fabric has ₹21,605.00(38.74%). The major reason for the loss caused by the discount. To clear the stocks and increase the sales of the product , they offered discounts. The men's unstitched fabric was sold 1148 units and purchased 20 units and left with 12 units . It generated gross sales of ₹2,39,145. The baby items were sold 138 units and purchased 140 and left with 20 units and generated gross sales ₹1,18,965.60.

Findings on monthly expenses and monthly gross profit :

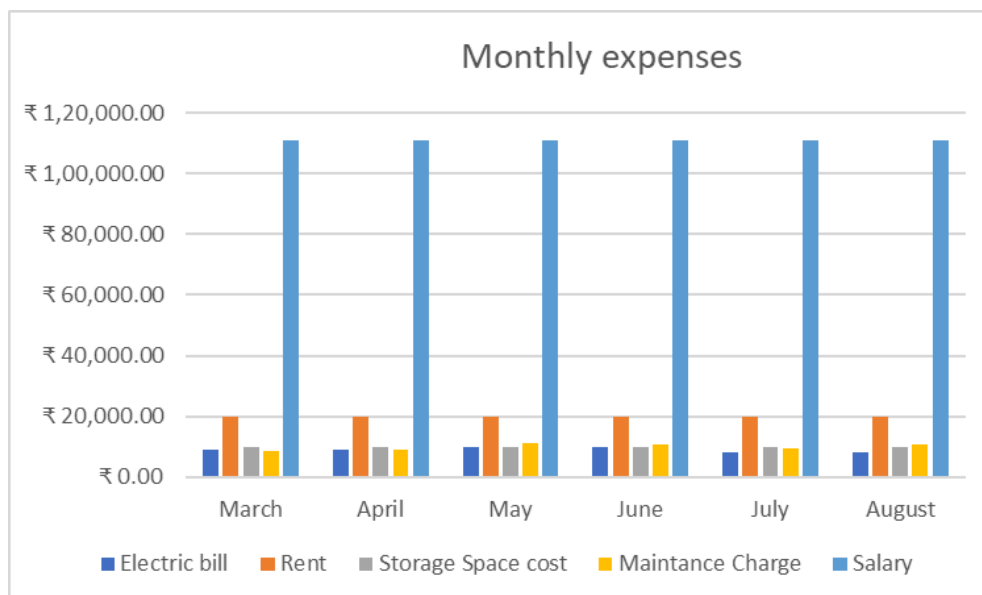


Chart-15 Clustered column chart showing the month-wise expenses.

The above bar chart shows that salary,rent,storage space cost remains unchanged for March-August. The maintenance charge and Electric bill shows slight fluctuations.

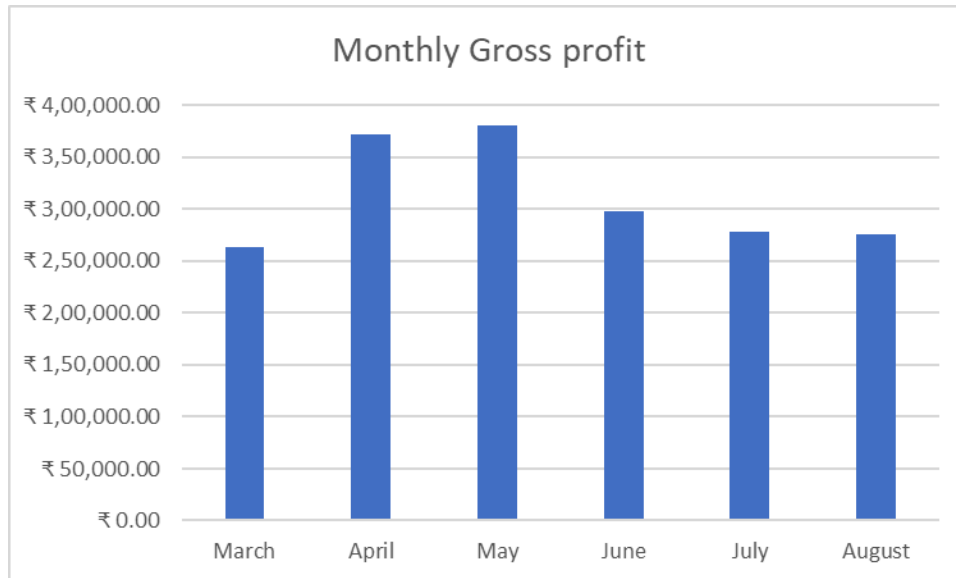


Chart-16 Column charts shows the monthly gross profit

On the other hand, The Monthly gross profit shows higher fluctuations . There was an incline of profit from March to April and April to May .

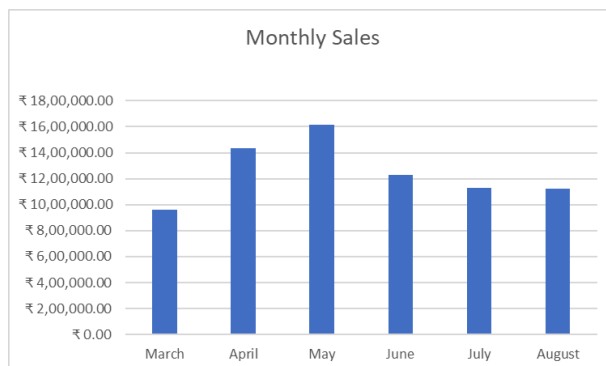


Chart-17 Column chart shows the monthly gross sales of all the products

This is due to the increase in the sales of April and May . After May there was a decline of sales and profit in the months of June,July,August.

If the expenses remain with minimal fluctuation and the gross profit keeps decreasing and increasing with huge variations this will affect the net profit.

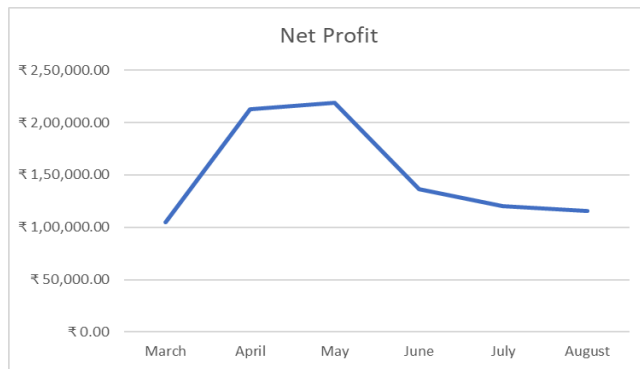


Chart-18 Line chart shows the net profit from all the products

If the net profit keeps decreasing it will cause financial problems . It reduces the cash flow , makes it difficult to invest , and hinders growth opportunities.

The below bar chart shows the profit earned by each product in the months of March-August.

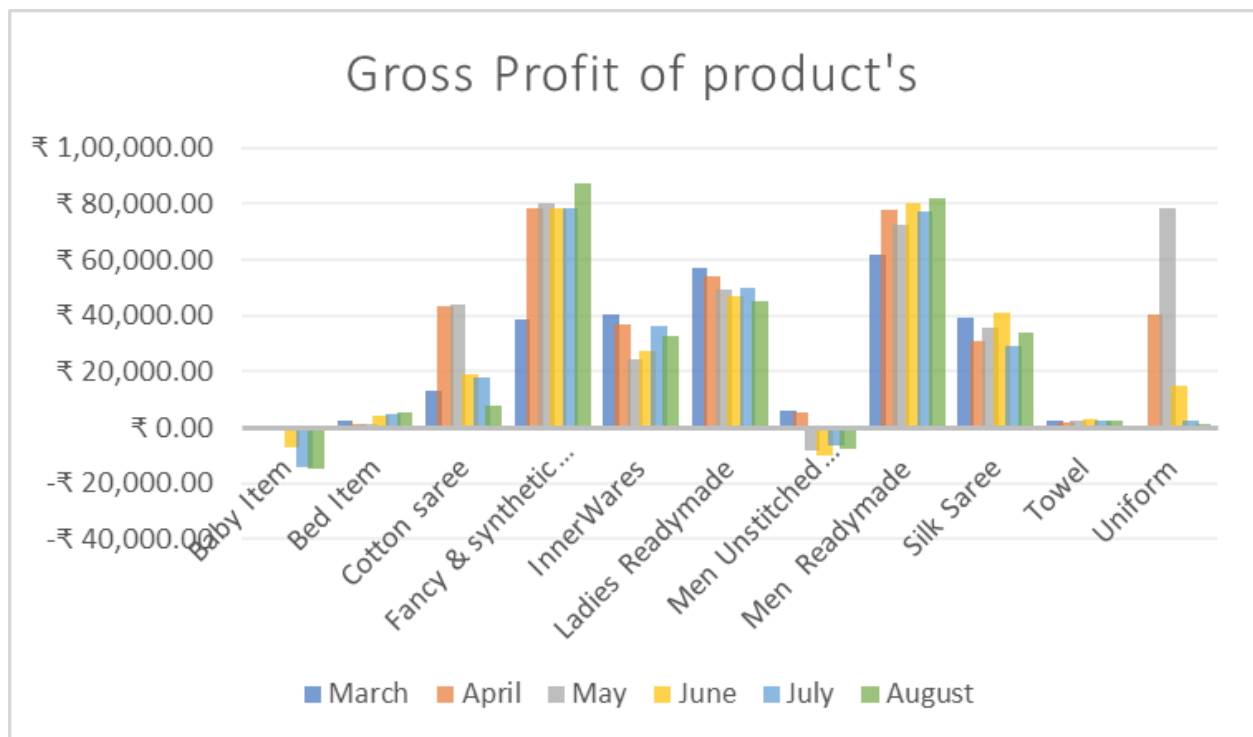


Chart-19 Clustered Column charts shows the Gross profit of product's across 6 months

Men readymade and Fancy&synthetic saree show significant gross profit across all the months. Ladies readymade and silk saree show moderate profit with moderate variability. Innerwares show moderate fluctuations in gross profit. The uniform and cotton shows a seasonal pattern. The gross profit of uniform was high during the May month and gross profit of cotton saree was higher in April and May than other months. The gross profit of bed items and Towel were lower throughout the March-August. The baby items generate loss in the months of June, July, August and show no noticeable profit in March, April and May. The men's unstitched fabric shows profit in March and April and loss in May, June, July, August.

INTERPRETATIONS OF RESULTS AND RECOMMENDATIONS

The Namnadu silks have been there for many years and MR. Muhamed Abbas Ali has implemented different things and improved the business. Here are the key insights and recommendations for further enhancement

Profit Analysis :

I have collected the data for 6 months (March-April) and performed several analyses. The net profit margin generated during those periods is 12.14% which is a moderate level of profitability for the textile industries. By implementing and following certain things, this profit can potentially increase. The correlation value of profit and sales is 0.96 which is very close to the 1 and indicates a positive strong relationship. So sales is the primary driver for profit. Increasing the sales of the product with marketing, promotional coupons can increase the profit.

Product-wise Recommendation :

Baby items :

Issues - It was sold with highest loss of 32.90% and item is overstocked in inventory

Action - Reduce purchase quantities and offer discounts of 15-20% can reduce the loss and gather customer feedback for this product and bring changes based on it

Bed items :

Issues - The gross profit for 6 months (₹19,827.60) and contribution to sales is low.

Action - Minimizing purchase quantities which avoid excessive stocks in the inventory.

Cotton Saree:

Issues - It has moderate profit and the demand is high in April and May.

Actions- Giving offers during the summer season and social media campaigns can increase level of sales. The stock level of cotton saree is maintained and it doesn't cause much loss to gross profit.

Men's Unstitched Fabric :

Issues - It has a loss of 9.03%. The demand is relatively low.

Action - Instead of increasing the discount percentage, Reducing the purchase quantities and giving some coupons for sales will help.

Towels :

Issues - It has lower sales and profit value but stable demand.

Action - Ensure enough stocks to avoid stockout during demand spikes.

Silk Sarees :

Issues - It has a higher gross profit margin of 31.03% but it has lower sales values. So overstocking the sarees affects the profit.

Action - Maintaining an appropriate stock level results in increase in profit.

Innerwares :

Issues- Profit and Sales are moderate and has a stable demand across all the months

Action - . As innerwares are fast moving products and don't depend on any seasonality, the incline in sales can contribute to the profit throughout the year. Introducing different brands can result in sale's incline.

Ladies Readymade :

Issues - Even Ladies readymade has higher sales value, the overstocking affects the profit from it. The ever changing trends in dressing style significantly impact the sales of ladies readymade if it is overstocked

Action - the purchasing method of ladies readymade can be changed. Instead of purchasing in bulk , they can purchase with limited amounts with respect to sales frequently . So that they can keep up with the changing trends.

Uniforms :

Issues : It has high sales in the month of May which is obviously due to the school reopening season.

Action - The Uniforms aren't need not be stored in other months in high quantities . It will affect the profit.They can plan to tie up with schools and institutions for bulk sales . It will increase sales on a larger level.

The men's readymade :

Issues : It has high sales,profit,balanced purchase and inventories.The demand of this product is throughout the year.

Action - Instead of investing more on men's unstitched clothes it is better to invest the cost on men's readymade clothes and expand its range. Varieties of models can be introduced in men's readymades. Ensure enough level of stock.

Fancy & synthetics saree :

Issues : It has higher sales , gross profit margin . The stocks and purchase is balanced with sales.It has stable demand

Action : The more sarees can be introduced with low maintenance , high durability , exclusive collections. Bundling offers can increase sales . Pair sarees with complimentary items like blouse,matching accessories will attract customers.During the festive season, the levels of stock have to be checked to meet the demand .

Additional Business Recommendations :

Expenses : The expenses are almost fixed and vary in a small amount. So the way to increase the profit is to boost sales and avoid overstocking.

Inventory management : Avoid overstocking of slow moving products like baby items,bed items.Focus on maintaining the stocks of products which have a stable demand .

Catching Display : Maintain an appealing display . Showcase the new arrival using focus light. Incorporate the seasonal themes in the display.This can attract new customers.

Training and skill development : They can hire a professional to train the workers and improve their skills and customer service.This can increase positive shopping experience to the customers and encourage them to return to the store.

Data Management : Following data management will result in growth of the business. Keeping the data in a detailed manner plays a vital role. Due to scarcity of data on daily consumptions , it is difficult to perform several analyses like reorder points. Performing those analyses will give an idea about inventory management. When the data is properly maintained on a daily basis that can be used to forecast demand ,plan inventories and satisfy customers needs.

Key Takeaways :

- Men's readymade, innerwares and Fancy&synthetic saree are the key products to improve sales.Prioritizing these product's sales and restocking the inventories whenever needed will help to increase profitability.
- Balance the inventories of seasonal products like uniform and cotton in the demand period and reduce the inventories of slow moving items like baby items,bed items,men's unstitched fabric.
- Reduce bulk purchase of ladies readymade to align with fashion trends.
- Focus more on marketing new products, promotions on online and data driven decision making

Incorporating several changes and following different strategies can help Namnadu silks to achieve sustainable development and long term growth .